Research GPT 5.6 Terra

Deep Third-Cumulant Representation for Random ReLU MLPs

Research response to Q1-Q3 based on the supplied benchmark note

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Executive conclusion

The supplied measurements rule out the obvious ways of making the deep third-cumulant tier cheap: dropping old sources, late reconstruction, memoryless slice closure, ordinary low-rank compression of Hadamard products, source averaging, scalar shared propagators, and tensor tricks that still require dense contractions. The remaining opportunity is therefore not to remove deep information, but to **change the state variable used to carry it**.

My main conjecture is to aggregate all old-source content into a single shared-basis state and compress that state in a representation whose rank is not the CP/Tucker rank of the individual Hadamard products. A tensor-train (TT) or closely related observable-state representation is the first mechanism I would test. A second route is to seek a low-dimensional quenched state-space realization of the mapping from deep third-cumulant state to the future observables D_{21} and D_3. A third, weaker route is an implicit matrix-vector oracle for the relevant contractions, followed by randomized low-rank recovery.

The key next measurement is an **aggregate observability-rank experiment**: estimate the minimal dimension of the old-source state needed to reproduce the future D_{21}/D_3 outputs to the required 2% RMS. This is the missing quantity that the current source-ablation experiments do not measure.

1 Problem constraints and what the measurements already establish

The benchmark is a width-1024, depth-16 random ReLU MLP with He initialization. The cost unit is one dense 1024×1024 matmul, equal to 2^{31} FLOPs; the budget is 1024 such units. The public reference points include covariance propagation at 24 units with raw MSE 4.05×10^{-6} and the V29 factorized $K=3$ chain at 260 units with raw MSE about 2.13×10^{-8}. Leading entries are reported at roughly 133-165 units and raw MSE around 1.64×10^{-8} to 1.9×10^{-8}.

The supplied analysis gives a decisive structural observation: the old-source legs are low-rank, but their elementwise products are near-full-rank and noise-like. At layers 6, 10, and 14, the legs themselves may need only roughly 100-260 effective directions, whereas products such as $A\circ P$, $A\circ A$, and $P\circ P$ need roughly 680-780 of 1024 directions for 99% energy.[1]

The later parity result is also important. For an old source, the source contribution is odd in the newest weight matrix while covariance and Gram information are even. Therefore the conditional expectation of the old-source third-cumulant contribution is exactly zero under the sign-symmetric weight ensemble. The noise-like Hadamard spectrum is thus not removable by an annealed average.

The sensitivity map closes the most tempting escape routes: late re-formation is inaccurate; every layer from approximately 8 onward materially matters; and no single birth is dispensable within the source-sum representation.

2 Q1 - Candidate formulations for the deep third cumulant

[1] All benchmark measurements in this report are taken as given from the supplied note.

2.1 1. Aggregate shared-basis TT / Hermite core

Status: Conjecture. Priority: highest.

Do not preserve ten old sources as ten independent (A_s, P_s) pairs. Instead form one aggregate old-state in the common basis $Q_c \in \mathbb{R}^{n\times r}$, $r = 384$, and represent its coefficient tensor as a tensor-train:

$$\kappa_3^{\text{old}} \approx (Q_c, Q_c, Q_c) \cdot T, \qquad T \in \mathbb{R}^{r\times r\times r}.$$

For TT rank ρ,

$$T_{abc} = \sum_{\alpha,\beta} G_1[a,\alpha] G_2[\alpha,b,\beta] G_3[\beta,c].$$

The crucial distinction from the already-dead CP/Tucker approach is that a TT core can have high CP rank while possessing moderate sequential unfolding ranks. The supplied measurements only show that generic Hadamard products have flat singular spectra; they do *not* measure TT ranks of the aggregate coefficient tensor.

A practical layer update is:

$$\widetilde{Q}_{l+1} = W^{(l+1)} D_{w_1^{(l)}} Q_l,$$

followed by re-orthogonalization, a basis change on the TT factors, addition of the new birth contribution, and TT truncation.

For the required observable,

$$D_{21,ic} = \sum_{a,b,d} Q_{ia} Q_{ib} Q_{cd} T_{abd}.$$

This contracts through the small factors and never forms dense A or P.

Estimated cost. With $n = 1024$, $r = 384$ and a candidate TT rank $\rho = 32$:

- propagating Q: about 0.375 units/layer,
- orthogonalization / basis maintenance: about $0.10 - 0.15$ units/layer,
- TT core transforms: about 0.15 units/layer,
- D_{21}/D_3 contractions: about 0.4 units/layer.

Thus the old tier would plausibly fall to roughly **1.1 units/layer**. At $\rho = 64$, a reasonable first estimate is roughly **2.5-2.8 units/layer**. These are analytical estimates, not measured benchmark results.

What it loses. Only TT truncation error, and the relevant criterion is induced D_{21} RMS error, not core Frobenius error alone. The target is $\lesssim 2\%$ relative RMS in D_{21}.

First falsifying experiment. Estimate TT ranks of the *aggregate old coefficient state* at one real network using randomized unfolding probes; test $\rho = 16, 32, 64, 128$. Reject the mechanism if $\rho \le 64$ cannot attain $< 2\%$ RMS error in D_{21}.

2.2 2. Observable-state / Hankel recursion

Status: Conjecture. Priority: high if the TT rank is not small.

A recursion using only D_{21}, D_3, and C is not enough in general because exact transport of a third-order tensor mixes in off-slice entries with unequal indices. The measured 18-23% missing slice energy in the dead memoryless closure is direct evidence that this remainder is substantial.

Instead define a small quenched hidden state $z_l \in \mathbb{R}^k$:

$$z_{l+1} = A_l z_l + B_l u_l, \qquad (D_{21}^l, D_3^l) = C_l z_l + D_l u_l.$$

Here z_l is a per-network state, not an annealed expectation. It should retain precisely the information that future D_{21} and D_3 observables can see.

The conceptual diagnostic is the numerical rank of a block-Hankel operator built from future slice responses to old-source perturbations. If that rank is small, a finite-dimensional state-space realization exists.

Estimated cost. A dense output map with state dimension k scales roughly as $O(kn^2)$, corresponding to

$$\frac{kn^2}{2^{31}} = 0.015625k \text{ units/layer.}$$

Thus $k = 32$ is roughly 0.5 units/layer before small state-update terms.

What it loses. The neglected tail of the observable Hankel singular spectrum.

First falsifying experiment. Build randomized block-Hankel observations from old-source perturbations over layers 8-16. Evaluate $k = 16, 32, 64, 128$. Reject the idea if even $k = 64$ leaves more than 2% RMS error in the future D_{21} observable.

2.3 3. Exact implicit matrix-vector oracle for $H = (A \circ P)A^T$

Status: Derived identity, but likely only a supporting mechanism.

For $A = QF_A$ and $P = QF_P$,

$$Hv = (A \circ P)A^T v.$$

Let

$$u = F_A^T Q^T v, \qquad S = F_A \operatorname{diag}(u) F_P^T.$$

Then exactly,

$$(Hv)_i = q_i^T S q_i.$$

Therefore a matrix-vector product can be evaluated without materializing either dense leg.

The asymptotic cost is approximately $O(nr^2)$ per vector. Numerically, with $n = 1024, r = 384$, one oracle application is roughly **0.28 units/vector/source**.

For ten old sources and k randomized probes, that is roughly $0.28 \times 10 \times k$ units/layer, before related contractions. Consequently this becomes attractive only if the *aggregate* observable has very small effective rank or if sources are merged first.

What it loses. Only the numerical low-rank approximation used after repeated matrix-vector products.

First falsifying experiment. Measure the singular-value tail of aggregate old D_{21}. If rank 32-64 does not give $< 2\%$ RMS error, abandon this route as a main mechanism.

2.4 4. Exact identity for a full cheap matrix?

Derived assessment: no generic $O(nr^2)$ full-matrix identity is apparent.

One can rewrite

$$H = \sum_{a,b} (q_a \circ q_b) c_{ab}^T Q^T,$$

where

$$c_{ab,c} = \sum_j F_{A,aj} F_{P,bj} F_{A,cj}.$$

The coefficient object is cubic in r, so forming it generically costs $O(nr^3)$, reproducing the arithmetic wall already observed experimentally. Hence the useful exact identity is for *matrix-vector products*, not for a cheap explicit matrix.

2.5 5. Shared propagator with per-source scalars

Status: weak conjecture; low priority.

A scalar model would require approximately

$$P_s \approx \alpha_s P_\star$$

in the directions that control D_{21}. The common downstream operator does not by itself imply scalar source alignment.

Cost. Potentially below 0.5 units/layer if valid.

First falsifier. Fit the optimal scalar α_s after projection into Q_c and measure the induced D_{21} residual. A residual above 2% immediately kills the mechanism.

3 Q2 - Three hypotheses for what the leaders carry

1. **Merged deep state independent of source count.** The cleanest explanation of the observed cost signature is that leaders keep young-source computations but collapse deep content into a shared state rather than paying once per old source-layer. A TT/Hermite state is one concrete implementation. *Observable:* total FLOPs versus number of old sources; near-flat scaling would support this.
2. **Low-dimensional quenched state-space recursion.** Leaders may carry a small hidden state whose dimension is tied to future observables rather than the number of births. *Observable:* a low numerical Hankel rank in the old-source channel.
3. **Implicit operator evaluation plus small-rank recovery.** Leaders may never form the relevant D_{21} operator explicitly, instead applying exact matrix-vector products and recovering only what the final contraction needs. *Observable:* very small effective rank of aggregate old D_{21} and cost scaling with probe count rather than source count.

The supplied public measurements do not identify which of these is actually used. The fact that leader cost is close to the V29 young-tier bill is a strong cost signature, but not a proof of any particular implementation.

4 Q3 - Single most informative next measurement

Measure aggregate observable state dimension.

For the combined old-source state, stack future observables D_{21} and D_3 across layers 8-16 into an observation operator $\mathcal{O}$. Construct rank-k approximations $K_{old}^{(k)}$ using randomized operator probes and evaluate

$$\epsilon_k = \frac{\|\mathcal{O}(K_{old}) - \mathcal{O}(K_{old}^{(k)})\|_F}{\|\mathcal{O}(K_{old})\|_F}, \qquad k \in \{16, 32, 64, 128\}.$$

Interpretation:

- $\epsilon_{32} < 2\%$: strong evidence for a cheap hidden-state or TT-style representation.
- $\epsilon_{64} < 2\%$ but $\epsilon_{32} > 2\%$: still highly promising; a few units/layer could plausibly replace the old tier.
- $\epsilon_{128} > 2\%$: low-state formulations become substantially less credible.
- Error remains large until near full dimension: the leaders probably exploit a different exact algebraic identity rather than compression.

This measurement is more informative than another birth-ablation or layer-ablation map. Those experiments already establish that the deep signal is needed; the missing variable is the **intrinsic dimension of the deep signal as seen by the future observables**.

Bottom line

Derived: the recorded measurements rule out averaging, omission, late reconstruction, memoryless slice closure, generic low-rank compression of the observed Hadamard products, and a generic cheap explicit full-matrix contraction.

Conjecture: the most promising unexplored mechanism is a *merged shared-basis representation* whose complexity is governed by TT / observable-state rank rather than by the rank of $A \circ P$ or by the number of old sources.

Measure next: determine the rank needed to reproduce the aggregate old-source D_{21}/D_3 response over the future layers. That single experiment should tell you whether the problem is fundamentally compressible in an observable-state sense, and it directly targets the unexplained gap between V29's 260-unit cost and the leaders' 133-165-unit cost.

Source basis: the supplied research note describing the 1024-wide, 16-layer random ReLU MLP benchmark, V29 measurements, sensitivity maps, parity result, and the listed dead approaches. Quantitative estimates in this response are analytical proposals unless explicitly marked as measured in the source note.